

Chapter 4 In-Class Exercise: Sorting Math Students

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1 Student Data

1. A placement test assigns each student a math score from 0 through 5. Begin with the following list of tuples:

```
students = [
    ("Maya", 5),
    ("Noah", 1),
    ("Olivia", 3),
    ("Liam", 4),
    ("Sophia", 2),
    ("Ethan", 0),
    ("Ava", 5),
    ("Lucas", 1),
    ("Emma", 3),
    ("Mateo", 4)
]
```

Each element of `students` is a tuple containing

(`student_name`, `math_score`)

For example,

```
print(students[0][0])
print(students[0][1])
```

prints the first student's name and score.

2 Course Rosters

1. Students are assigned according to

Score	Course
4, 5	Calculus
2, 3	Trigonometry
0, 1	Algebra

Create three course lists. Each list contains the course name followed by an initially empty student list:

```
algebra = [
    "Algebra", []
]

trigonometry = [
    "Trigonometry", []
]

calculus = [
    "Calculus", []
]
```

Combine the three lists into one tuple:

```
math_courses = (
    algebra,
    trigonometry,
    calculus
)
```

The final structure will therefore have the form

```
(
    ["Algebra", [students...]],
    ["Trigonometry", [students...]],
    ["Calculus", [students...]]
)
```

3 Conditional Sorting

1. Use conditional logic to place each student into the correct course. Begin with the first student:

```
name = students[0][0]
score = students[0][1]

if score >= 4:
    calculus[1].append(name)
elif score >= 2:
    trigonometry[1].append(name)
else:
    algebra[1].append(name)
```

The method

`.append(name)`

adds the student's name to the end of the appropriate student list.

- Repeat the conditional statement for `students[1]` through `students[9]`. Do not use a loop yet.
- For each student, extract the name and score first, then use one `if-elif-else` statement to make the placement.
- When all ten students have been processed, print `print(math_courses)` and verify that every student appears exactly once.

4 Check Your Result

1. Without changing the original `students` data, verify the following:
 - Students with scores 0 or 1 appear only in Algebra.
 - Students with scores 2 or 3 appear only in Trigonometry.
 - Students with scores 4 or 5 appear only in Calculus.
 - The three student lists contain a total of 10 names.